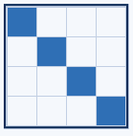
ORTHOGONALITY · PROJECTION FORMULAS

Four formulas, one operation

All four compute the same thing — the closest point of the target to $\mathbf{b}$. They differ only in what is assumed about the basis, and the formula gets simpler as the assumption gets stronger.

4
CASES

ONTO A SINGLE VECTOR — 1

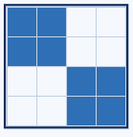


A line | the target is $\mathrm{Span}\{\mathbf{a}\}$

FORMULA	REQUIRES	COST PER PROJECTION
$\frac{\mathbf{b}\cdot\mathbf{a}}{\mathbf{a}\cdot\mathbf{a}}\mathbf{a}$	$\mathbf{a}\neq\mathbf{0}$	two dot products

The base case, and the one every formula below is built from. The denominator is what normalisation removes — for a unit vector it is 1 and disappears.

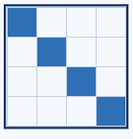
ONTO A SUBSPACE — BASIS ASSUMED ORTHOGONAL — 2



Orthogonal basis | $\mathbf{u}_i\cdot\mathbf{u}_j=0$ for $i\neq j$

FORMULA	REQUIRES	COST PER PROJECTION
$\sum\frac{\mathbf{b}\cdot\mathbf{u}_i}{\mathbf{u}_i\cdot\mathbf{u}_i}\mathbf{u}_i$	pairwise orthogonal, nonzero	$2k$ dot products

A sum of independent line projections. Orthogonality is what makes the terms independent — each coefficient can be computed without knowing the others, and adding a basis vector does not disturb them.

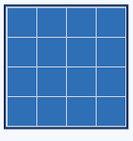


Orthonormal basis | orthogonal **and** $\|\mathbf{q}_i\|=1$

FORMULA	REQUIRES	COST PER PROJECTION
$\sum(\mathbf{b}\cdot\mathbf{q}_i)\,\mathbf{q}_i$	pairwise orthogonal, unit length	k dot products

The denominators vanish because each is 1. This is the form worth arranging for — see orthonormal operations, where every operation simplifies the same way.

ONTO A SUBSPACE — NOTHING ASSUMED — 1



Arbitrary basis | columns of A span the target

FORMULA	REQUIRES	COST PER PROJECTION
$A(A^{\mathsf{T}}A)^{-1}A^{\mathsf{T}}\mathbf{b}$	columns of A independent	a $k\times k$ inverse

The general case, and the expensive one. $(A^{\mathsf{T}}A)^{-1}$ exists exactly when the columns are independent — and when they happen to be orthonormal, $A^{\mathsf{T}}A=I$ and this collapses to the row above.

Reading upward is the argument for Gram–Schmidt. The general formula needs a matrix inverse; orthogonality removes the coupling between coefficients; normalisation removes the denominators. Each assumption bought by preprocessing is a computation avoided on every projection afterwards.